

DEV PARIKH

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SUMMARY

Software Engineer with 3+ years of experience building **scalable, production-grade AI systems and full-stack applications** across semiconductor and technology environments. Strong expertise in **LLM-based microservices, RAG pipelines, and deep learning models** using **Python, PyTorch, TensorFlow, LangChain, and FAISS**, optimized for **cloud and edge deployment**. Proven ability to deliver **FastAPI/Django backends, React frontends, and Spark-based data pipelines** on **AWS and Azure**, improving system performance and reliability. Hands-on experience with **CI/CD, Docker, Kubernetes, MLflow, and Prometheus/Grafana**, enabling high availability and efficient release cycles. Effective collaborator with a track record of translating complex requirements into **measurable, data-driven outcomes**.

SKILLS

Programming & Scripting: Python, R, SQL, Bash, JavaScript, TypeScript

Frontend Development: React.js, Next.js, Redux, HTML5, CSS3, Bootstrap, Tailwind CSS

Backend Development: Django, Flask, FastAPI, Node.js, Express.js, REST APIs, GraphQL

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, SQL Server, Snowflake, Delta Lake, Parquet

LLMs & Generative AI: Transformers, GPT Models, RAG Systems, Embeddings, Hugging Face, LangChain, Vector Databases (FAISS, Pinecone), Model Fine-Tuning, Prompt Engineering

Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Ensemble Methods, Time-Series Forecasting, Recommender Systems, Scikit-learn, XGBoost, LightGBM

Deep Learning: CNNs, RNNs, LSTMs, Autoencoders, GANs, Transfer Learning, PyTorch, TensorFlow, Keras

NLP & Computer Vision: BERT, spaCy, NLTK, NER, Text Classification, Topic Modeling, OpenCV, YOLO, Object Detection

Big Data & Distributed Computing: Apache Spark, PySpark, Spark MLlib, Hadoop, Hive, Kafka, Dask, Databricks

MLOps & Deployment: AWS (SageMaker, Lambda, EC2, ECR, S3), Azure ML, Docker, Kubernetes, MLflow, Airflow, Kubeflow, DVC, ONNX, TensorFlow Serving, TorchServe, CI/CD (GitHub Actions, Jenkins, GitLab CI), Prometheus, Grafana

Testing & Optimization: Unit Testing (PyTest, Jest), Postman, Swagger, A/B Testing, Model Drift Monitoring, Performance Tuning, Hyperparameter Optimization (Optuna, GridSearchCV, RandomizedSearchCV)

Analytics & Visualization: Power BI, Tableau, Matplotlib, Seaborn, Plotly

Version Control & Methodologies: Git, GitHub, GitLab, Agile, Scrum, SDLC

EXPERIENCE

Qualcomm, USA | *Software Engineer*

Jun 2025 – Present

- Designed and deployed **end-to-end LLM-powered microservices** using **FastAPI, Docker, and Kubernetes on AWS**, supporting production-grade inference at scale and reducing API response latency by **35%** across edge-device intelligence platforms.
- Built **RAG pipelines** using **LangChain, FAISS, and Hugging Face Transformers** to enable contextual semantic search over large technical knowledge bases, improving answer relevance and retrieval accuracy by **40%** in live environments.
- Optimized **CNN and transformer models** with **PyTorch and TensorFlow**, leveraging **ONNX and TensorRT** for model compression and acceleration, reducing model footprint by **50%** and enabling real-time inference on **Qualcomm edge hardware**.
- Developed and maintained automated **ML CI/CD pipelines** using **GitHub Actions and MLflow**, enabling continuous training, model versioning, and promotion across environments while reducing manual release cycles by **60%**.
- Collaborated closely with **product, platform, and research teams** to implement **model drift detection, monitoring, and controlled A/B experiments**, improving long-term production model stability and prediction quality by **25%**.
- Built **observable MLOps monitoring dashboards** using **Prometheus and Grafana** to track GPU utilization, inference latency, error rates, and API health, consistently meeting **99.5% uptime SLA** targets.
- Enhanced **large-scale data ingestion and feature pipelines** using **Apache Spark and Delta Lake**, improving data processing throughput by **45%** while increasing pipeline reliability and fault tolerance for ML workloads.

Mindtree, India | *Software Engineer*

Jan 2021 – Dec 2023

- Developed **full-stack analytics features** using **React.js, Django, and SQL** for a cloud-based enterprise analytics platform, reducing reporting turnaround time by **30%** and improving data accessibility for business stakeholders.
- Automated and scaled **data ingestion pipelines** using **PySpark and Apache Airflow**, improving data refresh reliability by **35%** and expanding integrations from **5 to 20+ upstream data sources**.
- Implemented and tuned **machine learning models** using **Scikit-learn and XGBoost** to support customer churn prediction use cases, increasing forecast accuracy from **72% to 85%** in production environments.
- Containerized backend services using **Docker** and enhanced deployment automation through **Jenkins-based CI/CD pipelines**, reducing manual deployment effort by **50%** and improving release consistency.
- Designed and delivered **interactive analytical dashboards** using **Power BI and Plotly**, enabling business teams to uncover actionable operational insights and improving data-driven decision-making efficiency by **40%**.
- Optimized **analytical SQL queries and data storage layers** using **Snowflake and PostgreSQL**, achieving up to **3× faster query performance** for large-scale reporting and analytics workloads.

EDUCATION

Master of Science in Data Science

University of Maryland, Baltimore County

Dec 2025

Baltimore, MD

Bachelor of Engineering in Computer Engineering

Gujarat Technological University

May 2023

Ahmedabad, India